

ANALYTICAL PROBLEM SOLVING – ASSESSMENT TOOL 1

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Course / Subject: Artificial Intelligence

1. Water Jug Problem

Given: A 4-gallon jug and a 3-gallon jug. Neither has measuring marks.

Goal: Obtain exactly 2 gallons of water in the 4-gallon jug.

State representation: (amount in 4-gallon jug, amount in 3-gallon jug)

<i>Step</i>	<i>Action</i>	<i>State</i>
1	Fill the 3-gallon jug.	(0, 3)
2	Pour the 3-gallon jug into the 4-gallon jug.	(3, 0)
3	Fill the 3-gallon jug again.	(3, 3)
4	Pour into the 4-gallon jug until it is full.	(4, 2)
5	Empty the 4-gallon jug.	(0, 2)
6	Pour the remaining 2 gallons into the 4-gallon jug.	(2, 0)

Result: Exactly 2 gallons are obtained in the 4-gallon jug.

2. Mars Rover – Intelligent Agent

i. Percepts: Rocks, soil, obstacles, temperature, atmospheric conditions, distance, terrain conditions, sample locations, and signs of water or minerals.

ii. Operating environment: Partially observable, dynamic, sequential, uncertain, continuous, and mainly single-agent.

iii. Actions: Move, turn, collect and analyze samples, take photographs, avoid obstacles, navigate, and communicate with Earth.

iv. Performance measures: Useful samples collected, accuracy of analysis, distance travelled, time taken, energy consumption, obstacle avoidance, scientific information obtained, and mission completion.

v. Suitable architecture: Model-Based Utility-Based Agent. It maintains information about the environment and chooses useful actions while considering goals such as sample collection, safety, and energy conservation.

3. 8-Queens Problem

Problem: Place 8 queens on an 8×8 chessboard so that no two queens attack each other.

Initial state: Empty chessboard.

State: A partial arrangement of queens on the chessboard.

Actions: Place a queen in a safe position in the next column.

Transition model: Adding a queen to a valid position produces a new state.

Goal test: All 8 queens are placed with no two queens in the same row, column, or diagonal.

Path cost: Each queen placement has a cost of 1; a complete solution has cost 8.

Suitable search: Backtracking / Depth First Search is commonly used because invalid arrangements can be rejected immediately.

One valid solution: [1, 5, 8, 6, 3, 7, 2, 4], where each number represents the row of the queen in successive columns.

4. OLA Cab Booking – Problem-Solving Agent

Problem: A customer wants to travel from a source to a destination using an OLA cab booking application. The customer may select Mini, Micro, Sedan, Prime, or Shared based on cost, comfort, and travel requirements.

Agent type: Utility-Based Agent.

Reason: The agent compares available choices using factors such as fare, travel time, comfort, and customer preference, then selects the option with the highest utility.

Pseudocode:

START

Input source location

Input destination

location Find available

cabs

For each available

cab: Calculate fare

Calculate travel time

Check comfort level

Calculate utility

Select cab with highest utility

Display selected cab, fare and estimated travel time

Ask customer for

confirmation If customer confirms:

Book the cab

Display booking details

Else:

Cancel booking

STOP

5. Logistics Delivery Network – Uniform Cost Search

Given edge costs: $S \rightarrow A = 1, S \rightarrow G = 12, A \rightarrow B = 3, A \rightarrow C = 1, B \rightarrow D = 3, C \rightarrow D = 1, C \rightarrow G = 2, D \rightarrow G = 3$. UCS

Uniform Cost Search (UCS): always expands the path having the lowest total cost so far.

Stage	Path / Node	Cost
Start	S	0
1	$S \rightarrow A$	1
2	$S \rightarrow A \rightarrow C$	2
3	$S \rightarrow A \rightarrow C \rightarrow D$	3
4	$S \rightarrow A \rightarrow C \rightarrow G$	4

Comparison: $S \rightarrow A \rightarrow C \rightarrow G$ has cost $1 + 1 + 2 = 4$, while the direct path $S \rightarrow G$ costs 12. Therefore, UCS selects the lower-cost route.

Final Answer: Least-cost path = $S \rightarrow A \rightarrow C \rightarrow G$; Minimum cost = 4.